

Anthony Colaiacovo

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TECHNICAL SKILLS

Programming Languages: Python, C++, C, JavaScript, SQL

Systems & Parallel Computing: Linux, CUDA, OpenMP, OpenCL, Metal, POSIX, Slurm

Web Technologies, Frameworks, & Databases: Flask, Electron, HTML, CSS, SQLite, PostgreSQL, PostGIS

Build & Deployment: Git, AWS, Docker, GitHub Actions, CMake, Makefile, SWIG

EXPERIENCE

Intermediate Software Developer

July 2026 – Present

Clearway Group of Companies

Vaughan, ON

- Owned the development, maintenance, and deployment of multiple internal applications relied upon daily by the pre-construction and estimating departments
- Developed a GIS-based haulage rate estimation application using the Google Maps API, OpenStreetMap data, and proprietary internal datasets to estimate trucking distances, routes, and haulage costs for construction projects
- Created and maintained an application that processes consolidated PDF invoices, splitting them by company division and adjusting pricing to produce the final invoices used for billing
- Mentored junior and intern developers through code reviews, technical guidance, and day-to-day collaboration

Junior Software Developer

Jan 2025 – July 2026

Clearway Group of Companies

Vaughan, ON

- Developed web and desktop applications for multiple internal projects using technologies such as Python, JavaScript, Flask, Electron, Pandas, PyMuPDF, PyQt, and RESTful APIs
- Built an automated system that processes incoming quote emails, identifies the relevant job and trade, and routes the quotes to the appropriate company directories, eliminating the need for manual email monitoring
- Created a program to identify, highlight, and measure vector data in construction drawings, reducing the time required to estimate material quantities such as pipe lengths
- Improved the job selection workflow for estimators by creating a tool that generates preliminary project statistics and presents insights through interactive dashboards built with JavaScript, HTML, and CSS

Undergraduate Research Assistant

May 2023 – Aug 2023

University of Guelph

Guelph, ON

- Optimized a GPU-accelerated biology simulation project written in C++ and OpenCL by refactoring the codebase, improving data access patterns within GPU kernels, and optimizing dynamic memory allocation and management, achieving up to 28% faster execution times
- Ported the program from OpenCL to CUDA and Metal, enabling performance comparisons across frameworks to identify the most efficient compute framework for different hardware architectures
- Automated the building and testing of the project with Python, eliminating the need for manual QA

Teaching Assistant

May 2022 – Apr 2023

University of Guelph

Guelph, ON

- Served as a teaching assistant for the University of Guelph's Web Design and Development (CIS1050), Programming (CIS1300), and Intermediate Programming (CIS2500) courses where I instructed students in C, HTML, and CSS
- Developed automatic grading software that enabled the TA team to finish grading in 35% of the allotted time
- Lectured on Git and GNU Make, led lab sessions, graded assignments and exams, and held office hours
- Selected to return as a teaching assistant for subsequent course offerings

PROJECTS

Warp (CIS4900 - Project write up available upon request) | C++, Python, LLVM, OpenMP, CMake

- Enhanced code generation in NVIDIA's Warp framework to enable parallel CPU execution through OpenMP directives, achieving up to a 4.7x speedup for CPU-bound workloads
- Configured LLVM and CMake to resolve OpenMP symbols during JIT compilation of dynamically generated code

EDUCATION

University of Guelph

Guelph, ON

Bachelor of Computing in Computer Science, Minor in Biology

Apr 2024

Notable Elective Topics: Systems Programming, Parallel Programming, Digital Systems, Computer Graphics